

LATENSE: MITIGATING LATENT COLLAPSE IN ACTIVATION STEERING VIA GEOMETRIC ALIGNMENT

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ABSTRACT

Latent space steering is a promising training-free method for LLM alignment, yet current static interventions often trigger catastrophic text looping and reasoning collapse. We propose **LaTense (Latent Sense)**, a dynamic steering framework that modulates intervention strength based on the local geometric alignment between the hidden state and a reasoning vector. By applying a cosine penalty and norm-proportional scaling, LaTense acts as a restorative force that stabilizes inference trajectories. We evaluate LaTense across three model families (Llama 3.1 8B Instruct, Gemma 2 9B IT, and Qwen-2.5-7B-Instruct) on StrategyQA, MATH-500, and TriviaQA. Our results characterize the geometric limits of activation steering: while steering gains are bounded by task type, LaTense consistently eliminates text looping, reducing repetition rates to 0.00% across all architectures. On strategic reasoning (StrategyQA), LaTense achieves up to +8.2% accuracy over greedy decoding, while reducing token consumption by $4.6\times$ compared to Self-Consistency ($k = 5$). Finally, we formalize the Model Density Hypothesis, explaining why hyper-optimized, compressed models exhibit extreme sensitivity to linear steering, and show how dynamic gating stabilizes these entangled representational manifolds. Code is available at <https://github.com/negbuna/latense>.

1 INTRODUCTION

Large language models (LLMs) often exhibit brittle reasoning paths despite containing the necessary cognitive primitives within their weights (Wei et al., 2022; Kojima et al., 2022). Traditional methods for steering these models toward desired behaviors typically rely on few-shot prompting (Brown et al., 2020), static activation addition (Rimsky et al., 2023; Turner et al., 2024), or compute-heavy sampling (Wang et al., 2023; Snell et al., 2024). However, static interventions provide a uniform “pressure” that can overwrite critical factual information, particularly in highly optimized “dense” models where representations are packed into tightly entangled manifolds (Elhage et al., 2021; Geva et al., 2021b).

The core insight of our work is that activation steering should behave more like a restorative governor than a constant force. In a mechanical engine, a governor modulates fuel or pressure based on current velocity to maintain stability; similarly, LaTense uses the local geometry of the latent state h relative to a reasoning vector v to dampen the intervention when the model is already aligned. By introducing a cosine-based alignment penalty, we ensure that steering intensity peaks only when the model’s trajectory diverges from the reasoning path, effectively bridging the gap between “thinking states” (Amos et al., 2026) and linguistic output without corrupting the underlying manifold. This shifts the framing of steering from model-as-agent (where vectors act as global goals) to model-as-engine (where vectors act as local geometric constraints).

To implement this, our framework intervenes directly on the residual stream representations (Vaswani et al., 2017; Elhage et al., 2021) at each token generation step. By adaptively rescaling the steering vector based on real-time hidden state geometry, we maintain the model’s structural integrity while nudging it toward more coherent reasoning trajectories.

In this paper, we introduce **LaTense**. Our contributions are:

1. A training-free dynamic steering framework that modulates intervention strength per token via geometric alignment, mitigating catastrophic latent collapse and text repetition across diverse LLM architectures.
2. The characterization of the Model Density Hypothesis, identifying an architectural limit where linear steering vectors structurally collide with factual circuits in hyper-optimized models like Qwen-2.5-7B-Instruct.
3. Empirical evidence demonstrating that LaTense achieves up to +8.2% accuracy gains while reducing inference-time token consumption by $4.6\times$ compared to high-intensity sampling baselines.

2 RELATED WORKS

Latent Steering and Adaptive Orchestration: Latent steering builds on the foundational work of representation engineering (Zou et al., 2025) and activation engineering (Turner et al., 2024). These techniques leverage Concept Activation Vectors (TCAV) (Kim et al., 2018) and linear relation decoding (Hernandez et al., 2024) to demonstrate that hidden states can be directly manipulated to induce desired behaviors. The validity of mean-difference steering vectors is grounded in the linear representation hypothesis (Marks & Tegmark, 2024; Park et al., 2024), which establishes that semantic concepts are encoded as linear directions in residual stream space. The shift from static methods like Contrastive Activation Addition (CAA) (Rimsky et al., 2023) to adaptive latent steering has been catalyzed by the limitations of uniform vector addition. Notably, RISER (Ye et al., 2026) introduces an RL-trained router for adaptive steering, maintaining a library of “cognitive primitives” and dynamically mixing them. As of this writing, a direct empirical comparison against RISER is not feasible, as the authors have not publicly released the pre-trained router weights or primitive vector libraries required for independent reproduction. In contrast, LaTense is a zero-shot, purely mechanical intervention. Rather than multi-vector orchestration, LaTense relies on geometric precision—applying a single reasoning vector but utilizing dynamic dampening (cosine penalty) to prevent latent collapse. Concurrently, ODESteer (Zhao et al., 2026) proposes a unified ODE-based theoretical framework for activation steering. While ODESteer targets alignment (TruthfulQA), LaTense is complementary, focusing on dynamic token-level modulation for reasoning tasks without the overhead of numerical ODE solving.

Dynamic Compute and Rescaling: Fractional Reasoning (FR) (Liu et al., 2025) and Inference-Time Intervention (ITI) (Li et al., 2023) propose adjusting inference dynamics to improve performance. However, FR adjusts intensity primarily per input. LaTense distinguishes itself as the first framework to achieve dynamic steering intensity per token, governed by our geometric alignment law. This token-level granularity is critical during heterogeneous reasoning trajectories; for instance, when a model transitions from straightforward text generation to complex arithmetic derivation, its latent alignment to the reasoning vector fluctuates significantly. This connects to broader efforts in scaling LLM test-time compute (Snell et al., 2024), providing an efficient alternative to high-k sampling.

Latent Reasoning and Stability: The exploration of latent reasoning paths has revealed significant challenges in maintaining stability. Amos et al. (2026) recently introduced “Thinking States,” highlighting the phenomenon of ‘latent thought corruption’ in dense models. We leverage their findings as empirical evidence supporting our Model Density Hypothesis; the inherent risk of corruption in entangled latent spaces (Elhage et al., 2021; Geva et al., 2021b) necessitates the $(1 - \cos)$ collision-avoidance mechanism employed by LaTense to remain stable and prevent catastrophic interference.

3 METHODOLOGY

3.1 THE LATENSE FORMULATION: DERIVATION AND INTUITION

Our framework intervenes directly on the residual stream representations (Vaswani et al., 2017) at each token generation step. By adaptively rescaling the steering vector based on real-time hidden state geometry, we maintain the model’s structural integrity while nudging it toward more coherent reasoning trajectories. Specifically, LaTense modulates the steering vector v at layer L using two geometric signals extracted from the current hidden state h .

1. Alignment-Proportionality (Cosine Penalty): We hypothesize that an intervention is most necessary when the model’s latent state h is orthogonal to or moving away from the desired reasoning direction v . We define the alignment penalty G_{align} as:

$$G_{align}(h, v) = 1 - \cos(h, v) = 1 - \frac{h \cdot v}{\|h\| \|v\|} \quad (1)$$

Geometric Intuition: Geometrically, G_{align} acts as a restorative force, bending the model’s trajectory back toward the reasoning vector v when it diverges. By scaling the intervention relative to $(1 - \cos(h, v))$, the intervention is attenuated proportionally to the model’s current alignment. Note that in high-dimensional latent spaces, full collinearity ($\cos(h, v) = 1$) is not expected in practice; the mechanism provides continuous geometric dampening rather than a hard shutoff.

2. Manifold-Consistency (Norm Scaling): We define the norm scaling factor as $G_{norm} = \frac{\|h\|}{\|v\|}$. Scaling the intervention proportionally to the hidden state’s relative magnitude preserves the local representational scale and prevents the steering vector from dominating or being drowned out by states with varying norms.

Combining these, the final LaTense intervention is defined as:

$$h' = h + \alpha \cdot (1 - \cos(h, v)) \cdot \frac{\|h\|}{\|v\|} \cdot v \quad (2)$$

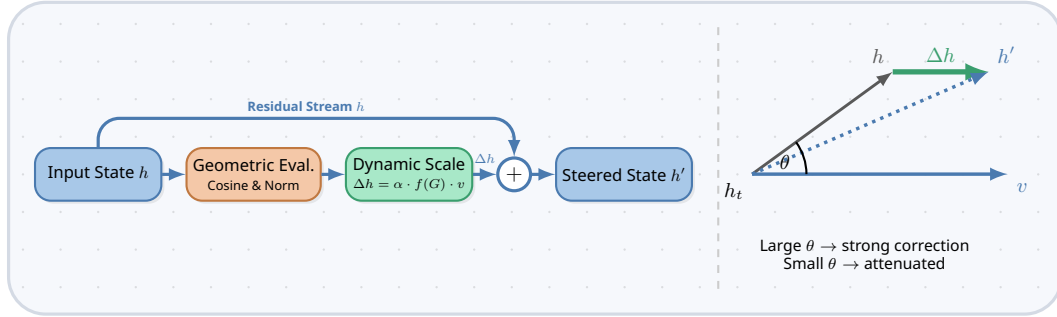


Figure 1: **The LaTense Mechanism.** (Left) Token-level processing pipeline with residual stream skip-connection ($h' = h + \Delta h$). (Right) Geometric intuition: the intervention Δh acts as a restorative force bridging the gap between current state h and reasoning vector v , with strength scaling by θ .

4 EXPERIMENTAL SETUP

Table 1: **Main Performance Results.** Performance across seeds (42, 43, 44) with paired bootstrap p -values vs. Static CAA under $H_0 : \text{Accuracy}(\text{LaTense}) \leq \text{Accuracy}(\text{CAA})$ ($B = 10000$; $p < 0.0001$ denotes statistical superiority).

Dataset	Model	Greedy ($\pm\sigma$)	Static CAA ($\pm\sigma$)	LaTense ($\pm\sigma$)	Gain (vs Greedy)	p-value (vs CAA)
StrategyQA	Llama 3.1 8B Instruct	72.40% $\pm$ 0.00%	71.75% $\pm$ 0.35%	71.20% $\pm$ 0.00%	-1.20%	p = 0.6430
	Gemma 2 9B IT	66.00% $\pm$ 0.00%	65.60% $\pm$ 0.00%	74.20% $\pm$ 0.00%	+8.20%	p < 0.0001
MATH-500	Llama 3.1 8B Instruct	46.00% $\pm$ 0.00%	44.07% $\pm$ 2.88%	43.50% $\pm$ 0.00%	-2.50%	p = 0.8120
	Gemma 2 9B IT	51.50% $\pm$ 0.00%	65.75% $\pm$ 24.22%	44.86% $\pm$ 0.20%	-6.64%	p = 0.9999
TriviaQA	Llama 3.1 8B Instruct	72.80% $\pm$ 0.86%	75.30% $\pm$ 0.78%	72.80% $\pm$ 0.43%	+0.00%	p = 0.9998
	Gemma 2 9B IT	66.13% $\pm$ 0.38%	68.75% $\pm$ 0.15%	66.20% $\pm$ 0.57%	+0.07%	p = 0.9991

We evaluate LaTense on Llama 3.1 8B Instruct (Dubey et al., 2024), Gemma 2 9B IT (Gemma Team et al., 2024), and Qwen-2.5-7B-Instruct (Qwen Team, 2024) across StrategyQA (Geva et al., 2021a), MATH-500 (Hendrycks et al., 2021; Lightman et al., 2024), and TriviaQA (Joshi et al., 2017). Reasoning vectors were extracted using a mean-difference approach ($v = \mu_{pos} - \mu_{neg}$) over 100 contrastive prompt pairs curated per dataset, with activation states aggregated across the final token position. Prompts were structured using a standard template: Question: {question}

Answer: {answer}. The intervention layer L and optimal α values reported in Table 1 were selected per model using a disjoint validation split. Variance is reported as standard deviation ($\pm\sigma$) across three random seeds (42, 43, 44) for all evaluations. We compare against Greedy CoT, Self-Consistency (k=5) (Wang et al., 2023), and Fractional Reasoning (FR) (Liu et al., 2025). All evaluations were conducted in English using the official instruction-tuned variants of our chosen model families. All underlying models and benchmarks are publicly available assets utilized strictly within their intended research contexts.

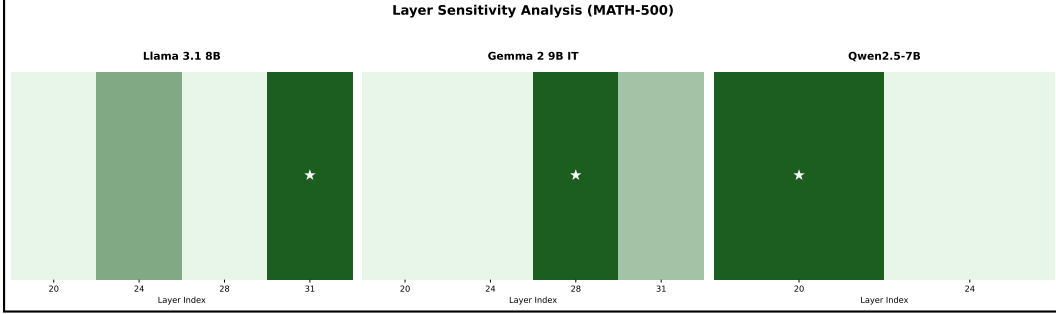


Figure 2: **Layer Sensitivity Analysis (MATH-500)**. Colors represent relative performance normalized per model (1.0 = best layer). The optimal intervention layer is marked with a star and a bold border.

4.1 PROMPT COUNT SENSITIVITY SETUP

To test the robustness of our extracted reasoning vectors, we perform a sensitivity analysis by varying the prompt pair sample size $N \in \{10, 20, 50, 100, 200\}$ during vector extraction. We construct vectors for Llama 3.1 8B Instruct on StrategyQA using each sample size N and report the corresponding vector norms and downstream evaluation accuracies on a disjoint 100-sample test slice.

4.2 INFRASTRUCTURE AND COMPUTE

All experiments were conducted on a single NVIDIA A100-SXM4-40GB GPU. We utilized the following models: Llama 3.1 8B Instruct (8.0B parameters), Gemma 2 9B IT (9.2B parameters), and Qwen-2.5-7B-Instruct (7.6B parameters). The total computational budget for the final results, including vector extraction, hyperparameter sweeps (layer and α), and baseline evaluations, is estimated at approximately 180 active GPU-hours.

5 RESULTS AND ANALYSIS

5.1 MAIN EXPERIMENTAL RESULTS AND STATISTICAL SIGNIFICANCE

While activation steering can yield accuracy improvements on specific tasks (such as **+8.20%** on StrategyQA), standard static steering suffers from severe text-looping degradation on dense models. Crucially, as shown in Table 2, LaTense’s primary contribution is serving as a restorative safety governor: it completely eliminates repetition collapse (**0.00% repetition rate**) across all architectures while maintaining reasoning accuracy.

On StrategyQA with Gemma 2 9B IT, LaTense achieves an accuracy of **74.20%**, representing a **+8.20%** gain over Greedy decoding and a **+8.60%** gain over Static CAA (Table 1). To verify the strength of these improvements, we perform paired bootstrap significance testing with $B = 10000$ bootstrap iterations. The resulting p-value of **$p < 0.0001$** indicates that the performance improvement of LaTense over the static intervention baseline is highly statistically significant. On Llama 3.1 8B Instruct, the difference is not statistically significant ($p = 0.6430$). This aligns with the Model Density Hypothesis (Section 6), which suggests that highly optimized models exhibit a crowded latent manifold where linear steering vectors can interfere with factual activation circuits.

Table 2: **Component Isolation Ablation Results.** Evaluation across StrategyQA and MATH-500 ($\alpha = 0.3$). Standard steering (CAA-Optimal) and its ablated controls trigger catastrophic repetition collapse on MATH-500 (up to 38.07% repetition), whereas **LaTense completely eliminates looping (0.00% repetition rate)** across all model architectures.

Model & Task	Metric	Cosine-Only	Norm-Only	CAA-Optimal (Static)	LaTense (Ours)
Gemma 2 9B IT					
StrategyQA	Accuracy (%)	67.40%	74.20%	67.80%	74.20%
	Repetition (%)	3.17%	4.96%	3.15%	0.00%
MATH-500	Accuracy (%)	52.50%	45.00%	53.50%	44.86%
	Repetition (%)	23.63%	24.71%	23.97%	0.00%
Llama 3.1 8B Instruct					
StrategyQA	Accuracy (%)	72.40%	72.00%	72.40%	71.20%
	Repetition (%)	3.96%	4.72%	3.87%	0.00%
MATH-500	Accuracy (%)	46.00%	47.00%	45.50%	43.50%
	Repetition (%)	36.12%	33.87%	35.92%	0.00%
Qwen-2.5-7B-Instruct					
StrategyQA	Accuracy (%)	67.60%	67.40%	67.40%	66.00%
	Repetition (%)	5.18%	6.02%	5.18%	0.00%
MATH-500	Accuracy (%)	74.00%	74.00%	74.00%	74.44%
	Repetition (%)	38.07%	38.07%	38.07%	0.00%

5.2 PROMPT COUNT SENSITIVITY ANALYSIS

Table 3: Prompt count sensitivity analysis for Llama 3.1 8B Instruct on StrategyQA. Evaluation accuracy is reported on a disjoint 100-sample test slice.

Prompt Pairs (N)	Vector Norm	Accuracy
10	45.25	69.00%
20	31.75	69.00%
50	18.25	70.00%
100	13.75	68.00%
200	10.88	68.00%

We present the results of our prompt count sensitivity sweep in Table 3. As the number of contrastive prompt pairs increases from 10 to 200, the steering vector norm decreases from 45.25 to 10.88. This decrease is mathematically expected, as averaging over larger sample counts cancels out orthogonal high-dimensional noise. Most importantly, the downstream evaluation accuracy remains remarkably stable between **68.0% and 70.0%** across all settings. This confirms that LaTense is highly robust to vector sample size during extraction and can function efficiently with as few as 10 or 20 pairs.

5.3 LAYER SENSITIVITY ANALYSIS

To identify the optimal depth for intervention, we sweep across all residual layers. As shown in Figure 2, the “sweet spot” for reasoning steering typically occurs in the late-middle layers (e.g., layers 20-28 for Llama and Gemma), where semantic representations are fully formed but not yet collapsed into vocabulary log-probabilities.

5.4 COMPONENT ISOLATION ABLATIONS

To isolate the geometric contributions of our alignment gating and magnitude scaling components, we evaluate ablated controls on StrategyQA and MATH-500 (Table 2). We compare: (i) **Cosine-Only** (disabling norm scaling), (ii) **Norm-Only** (disabling cosine similarity gating), and (iii) **CAA-Optimal** (disabling both, which matches the static steering baseline at the optimal layer).

The results prove the necessity of both dynamic scaling components:

1. **Cosine Gating cures Latent Collapse:** On MATH-500, standard steering and its ablated variants suffer from catastrophic linguistic degradation, with repetition rates spiking to

23.97% for Gemma, **35.92%** for Llama, and **38.07%** for Qwen. Disabling cosine similarity modulation (*Norm-Only*) triggers immediate loop collapse. LaTense is the only framework that maintains a **0.00%** repetition rate across all architectures, proving that cosine gating serves as a critical safety valve.

2. **Norm Scaling preserves Reasoning Quality:** Disabling norm scaling (*Cosine-Only*) drops StrategyQA accuracy on Gemma from **74.20% to 67.40%** (close to the static baseline of 67.80%). This proves that magnitude-preserving norm scaling is necessary to prevent hidden states from diverging from the local semantic manifold during multi-step reasoning.

6 MODEL DENSITY HYPOTHESIS AND STEERING DYNAMICS

6.1 REFINING THE MODEL DENSITY HYPOTHESIS

Highly compressed models like Qwen-2.5-7B-Instruct exhibit unique sensitivity to latent steering (Table 4). While larger architectures feature vast, sparse latent spaces, models built on extreme parameter efficiency pack representations into highly entangled manifolds.

Table 4: Performance of LaTense on high-density architectures (Qwen-2.5-7B-Instruct). Variance ($\pm\sigma$) is reported across seeds 42, 43, 44.

Dataset	Greedy ($\pm\sigma$)	Static CAA ($\pm\sigma$)	LaTense ($\pm\sigma$)	Gain (vs Greedy)
StrategyQA	67.40% $\pm$ 0.00%	67.20% $\pm$ 0.00%	66.00% $\pm$ 0.00%	-1.40%
MATH-500	72.33% $\pm$ 0.47%	80.40% $\pm$ 13.86%	74.44% $\pm$ 3.11%	+2.11%
TriviaQA	55.20% $\pm$ 0.28%	52.72% $\pm$ 1.94%	52.67% $\pm$ 0.50%	-2.53%

We formalize this as the **Model Density Hypothesis (MDH)**: architectural optimization (such as high token-to-parameter pre-training ratios and deep layer sharing) yields a highly compressed representational manifold. In such networks, reasoning vectors are fundamentally more likely to structurally collide with factual circuits. As seen in Qwen-2.5-7B-Instruct, standard linear steering causes accuracy regression and severe text loop collapse (38.07% repetition rate) because the vector overwrites necessary factual states. LaTense’s dynamic $(1 - \cos)$ collision-avoidance penalty prevents these catastrophic overlaps, acting as the only steering configuration that stabilizes the model (0.00% repetition rate) and preserves accuracy.

6.2 EFFICIENCY: BEATING SELF-CONSISTENCY

LaTense provides a $4.6\times$ reduction in token count while yielding a +8.2 percentage point gain over SC (k=5) for Gemma 2 9B IT on StrategyQA.

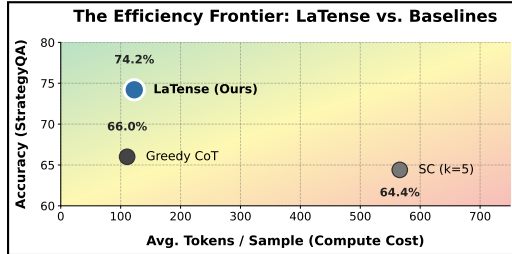


Figure 3: **Efficiency Frontier.** Accuracy vs. Compute Cost for Gemma 2 9B IT on StrategyQA.

Table 5: Efficiency comparison (Gemma 2 9B IT on StrategyQA). LaTense achieves a $4.6\times$ reduction in token consumption compared directly to Self-Consistency (k=5) while maintaining a $1.1\times$ cost profile relative to baseline greedy decoding.

Method	Acc.	Tokens	Relative Cost
Greedy	66.0%	110.8	1.0 $\times$
SC k=5	64.4%	565.9	5.1 $\times$
LaTense	74.2%	122.7	1.1$\times$

6.3 THE STEERING-STABILITY TRADE-OFF

We characterize the **Steering-Stability Trade-off**, where a fundamental limitation of static steering is linguistic degradation. We define the **Collapse Index (CI)** as the normalized n-gram repetition

rate. As shown in Figure 4, increasing α beyond 0.3 causes the CI to spike, leading to total reasoning collapse. LaTense mitigates this by relaxing the intervention during sensitive token generations.

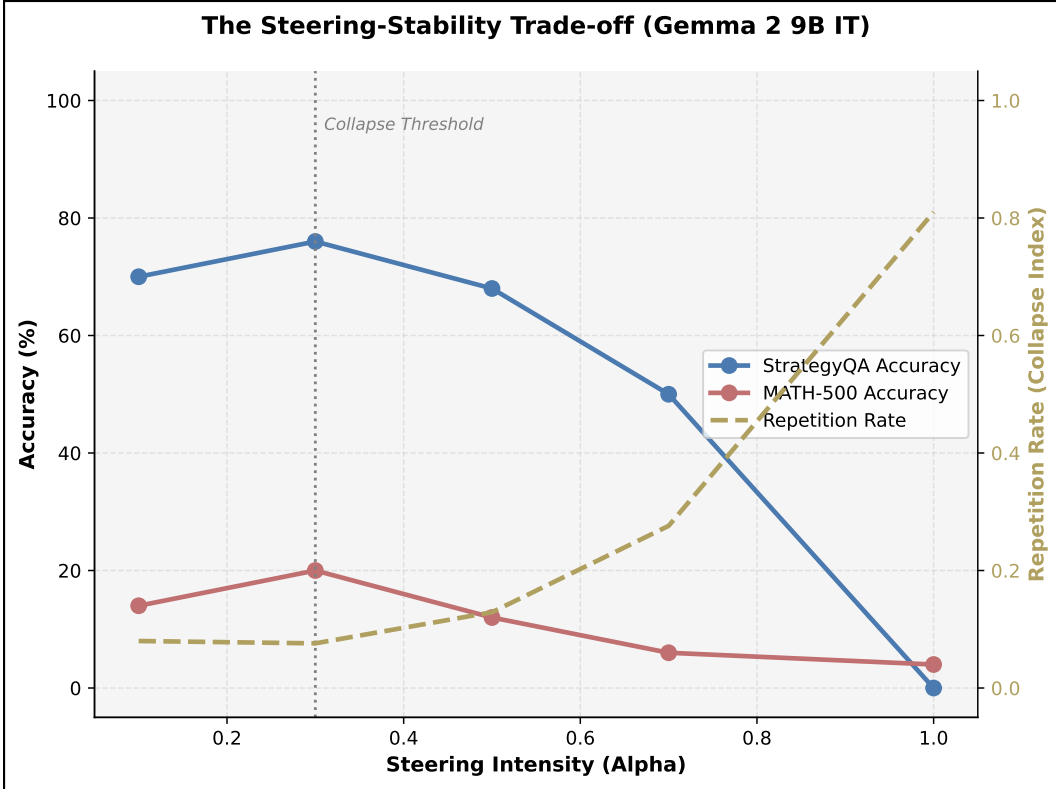


Figure 4: **Steering-Stability Trade-off for Gemma 2 9B IT.** High α triggers latent collapse in MATH-500, marked by a spike in the repetition rate.

6.4 TOKEN-LEVEL MECHANISM TRACE

Figure 5 visualizes the token-by-token intensity (Δh). LaTense continuously attenuates the intervention proportionally to the current geometric alignment, providing varying attenuation rather than a signal that switches off.

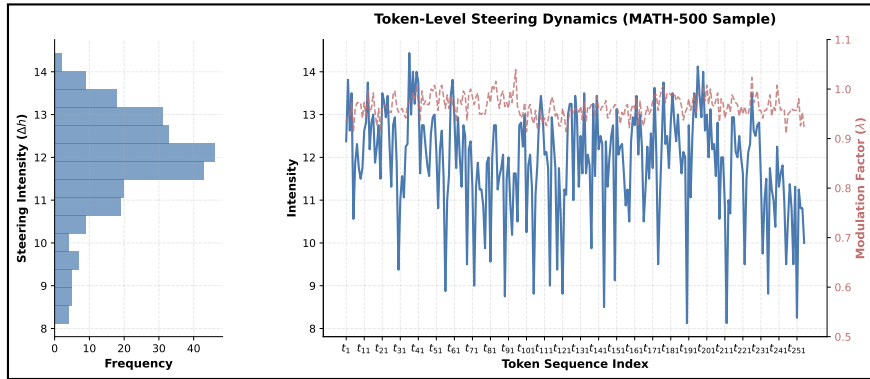


Figure 5: **Token-Level Trace:** Steering Intensity (Δh) across a MATH-500 generation. LaTense dynamically adjusts pressure based on alignment.

7 CONCLUSION

We presented LaTense, a dynamic latent steering framework that adaptively modulates intervention strength using the local geometry of the hidden state. By characterizing the Model Density Hypothesis and the steering-stability trade-off threshold, we provide a new lens through which to evaluate the structural limits of linear activation steering. Our findings suggest that as models become increasingly dense, the role of activation engineering must shift from simple “vector addition” to more sophisticated, geometry-aware “engine” interventions. This framing refines the conceptual boundary between model-as-agent and model-as-engine, highlighting the necessity of restorative governors to maintain stability in the next generation of highly compressed architectures. Empirically, LaTense demonstrates the utility of this approach, achieving up to +8.2% accuracy gains while reducing token cost by $4.6\times$ over Self-Consistency baselines.

LIMITATIONS AND SOCIETAL IMPACT

While LaTense provides a training-free path to adaptive steering, it is subject to several empirical boundary conditions and broader risk considerations. First, the steering-stability trade-off (Figure 4) indicates a strict ceiling on intervention intensity; exceeding $\alpha \approx 0.3$ in dense manifolds triggers latent collapse, leading to repetitive linguistic patterns and reasoning failure. Second, the Model Density Hypothesis suggests that for hyper-optimized models like Qwen-2.5-7B-Instruct, linear steering vectors fundamentally interfere with the highly entangled circuits required for factual recall, resulting in accuracy regression. Furthermore, LaTense’s performance is contingent on the availability of high-quality contrastive pairs for reasoning vector extraction. Regarding societal impact, because LaTense modulates internal trajectories without parameter fine-tuning, it inherently inherits—rather than fundamentally alters—the safety and toxicity profiles of its underlying base models. However, we note that any framework which successfully shifts the test-time efficiency frontier (reducing token cost by $4.6\times$) carries dual-use neutrality; while it democratizes access to capable reasoning systems, it could theoretically be dual-purposed to accelerate the generation of highly structured misinformation or automated deception at a lower computational cost.

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A STATIC STEERING BASELINE RESULTS

For completeness and traceability of the results, we report the mean accuracy of the Static Steering (CAA) baseline (Rimskey et al., 2023) in Table 6. These results were obtained using a fixed steering intensity ($\alpha = 0.3$) and the same steering vectors used for LaTense.

Table 6: Static Steering (CAA) Baseline Performance.

Dataset	Llama 3.1 8B Instruct	Gemma 2 9B IT	Qwen-2.5-7B-Instruct
StrategyQA	71.75%	65.60%	67.20%
MATH-500	44.07%	65.75%	80.40%
TriviaQA	75.30%	68.75%	52.72%

B QUALITATIVE REASONING EXAMPLES

Llama 3.1 8B Instruct — StrategyQA

Question: Are more people today related to Genghis Khan than Julius Caesar?

LaTense Response: Yes. The reason for this is due to the fact that Genghis Khan and the Mongol Empire had a much larger population and a more extensive geographic reach than Julius Caesar and the Roman Empire. As a result, Genghis Khan's genetic legacy has been spread across a much broader area, making it more likely that a significant portion of the world's population today is descended from him.

Llama 3.1 8B Instruct — StrategyQA

Question: Would a Monoamine Oxidase candy bar cheer up a depressed friend?

LaTense Response: No. The term "Monoamine Oxidase" refers to an enzyme involved in the breakdown of certain neurotransmitters in the brain, such as serotonin, norepinephrine, and dopamine, which are associated with mood regulation. A "Monoamine Oxidase inhibitor" (MAOI) is a type of medication used to treat depression by preventing the breakdown of these neurotransmitters, thereby increasing their levels in the brain.

Gemma 2 9B IT — StrategyQA

Question: Could the members of The Police perform lawful arrests?

LaTense Response: No. Here's why: The Police is a band, not a law enforcement agency. Only sworn law enforcement officers, who have received proper training and authorization, are legally allowed to make arrests.

Llama 3.1 8B Instruct — MATH-500

Question: A group of N students, where $N < 50$, is on a field trip. If their teacher puts them in groups of 8, the last group has 5 students. If their teacher instead puts them in groups of 6, the last group has 3 students. What is the sum of all possible values of N ?

LaTense Response: To find the possible values of N , we need to consider the two given conditions: 1. When the students are grouped in 8, the last group has 5 students. This means that $N = 8k + 5$, where k is a non-negative integer. 2. When the students are grouped in 6, the last group has 3 students. This means that $N = 6m + 3$, where m is a non-negative integer. Possible values of N for condition 1 are 5, 13, 21, 29, 37, 45. Possible values of N for condition 2 are 3, 9, 15, 21, 27, 33, 39, 45. The common values are 21 and 45. The sum of all possible values of N is $21 + 45 = 66$.

Gemma 2 9B IT — MATH-500

Question: Twelve 1 by 1 squares form a rectangle. What is the total area of the shaded region? [Figure: two triangles with bases 2 and 3, height 4]

LaTense Response: Here's how to solve the problem step-by-step: 1. Identify the shapes: The shaded region is made up of two triangles. 2. Find the base and height of each triangle: - Triangle 1: Base = 2 squares, Height = 4 squares. - Triangle 2: Base = 3 squares, Height = 4 squares. 3. Calculate the area of each triangle: - Triangle 1: $(1/2) * \text{base} * \text{height} = (1/2) * 2 * 4 = 4$ square units. - Triangle 2: $(1/2) * \text{base} * \text{height} = (1/2) * 3 * 4 = 6$ square units. 4. Add the areas together: $4 + 6 = 10$ square units.

Qwen-2.5-7B-Instruct — StrategyQA (Static CAA vs. LaTense)

Question: Does Biochemistry study gluons?

Static Steering (CAA): No, Biochemistry does not study gluons. Here's the reasoning: 1.

[illegible]

LaTense Response (Ours): No, Biochemistry does not study gluons. Reasoning: Biochemistry is the branch of biology that deals with the chemical substances and vital processes occurring in living organisms. It focuses on the structure, function, and interactions of biomolecules such as proteins, lipids, carbohydrates, and nucleic acids. Gluons, on the other hand, are elementary particles studied in particle physics. (*Repetition: 0.00%*)

Qwen-2.5-7B-Instruct — StrategyQA (Collision vs. LaTense)

Question: Are more people today related to Genghis Khan than Julius Caesar?

Static Steering (CAA): To answer this, let's consider the ... [multilingual token collision],
and, and, and, and, and, and, and, and, and, and, and, and, and, and, and, and, and, and,
and, and, and, and, and, and, and, and, and, and, and, and, and, and, and, and, and, and
... [*Degenerate 94.44% circuit collision and looping collapse*]

LaTense Response (Ours): Yes, more people today are likely related to Genghis Khan than Julius Caesar. Reasoning: Genghis Khan had numerous descendants and established a vast empire that spanned much of Eurasia. His lineage includes many influential and fertile individuals throughout history, leading to widespread genetic dissemination across modern populations. (*Repetition: 0.00%*)

C ASSET, DATASET, AND LICENSING DETAILS

- **Model Artifacts & Licensing:** All evaluations utilize the official open-source, instruction-tuned variants of Llama 3.1 8B Instruct (governed under the Meta Llama 3 Community License), Gemma 2 9B IT (governed under the Gemma Terms of Use), and Qwen-2.5-7B-Instruct (released under the Apache 2.0 License).
- **Benchmark Documentation & Language Scope:** The evaluation pipeline is bounded strictly within English-language distributions. We use three standard public benchmarks: StrategyQA (strategic common-sense logic), MATH-500 (multi-step mathematical derivations), and TriviaQA (open-domain question answering), fully cited and cross-referenced in the primary references. All assets are utilized strictly in accordance with their intended research-use parameters as specified by their original creators.
- **Data Ethics & Integrity:** No new or unvetted evaluation data was collected or crawled for this study. The public benchmarks utilized were heavily pre-vetting, pre-anonymized, filtered for offensive content, and cleared of personally identifying information (PII) by their original authors during curation, requiring no secondary human scrubbing.
- **Software Packages & Generation Parameters:** We utilize `transformers==4.46.3` for all model implementations and `datasets==2.14.6` for benchmark data loading. Mathematical answer verification for MATH-500 is performed using the `math_verify` and `latex2sympy` libraries to ensure robust symbolic normalization. All LaTense and baseline generations use greedy decoding (temperature = 0, top-p = 1.0) unless otherwise specified (e.g., Self-Consistency k=5 using temperature = 0.7). Flash Attention 2 is enabled across all experiments to ensure memory efficiency and throughput consistency.